



Cloud Compute Solution Design



Foreword

- This lesson describes Elastic Cloud Server (ECS) features, instance selection, initialization mode selection, and storage solution selection, as well as the definitions, features, and application scenarios of Bare Metal Server (BMS) and Dedicated Host (DeH). This lesson also discusses the five principles of computing architecture.



Objectives

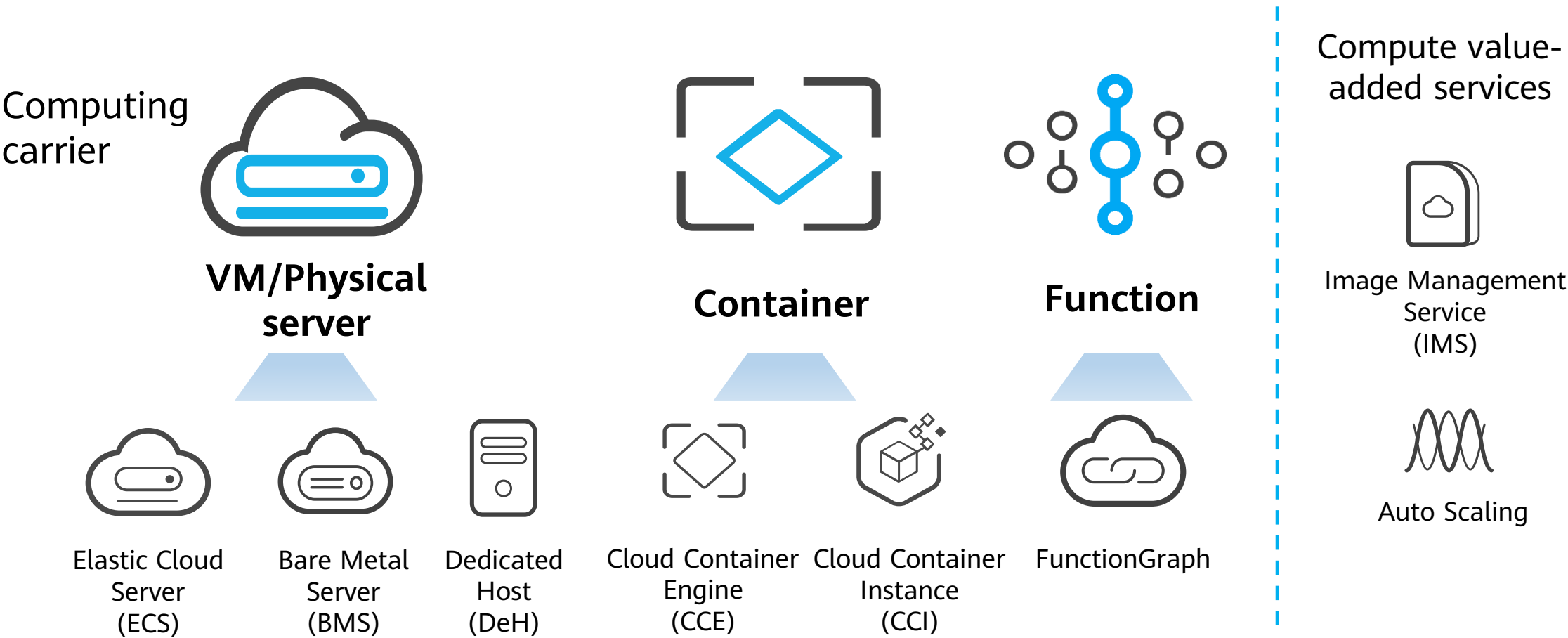
- Upon completion of this course, you will:
 - Understand basic Huawei Cloud compute services
 - Be able to optimize the computing architecture based on the five principles



Contents

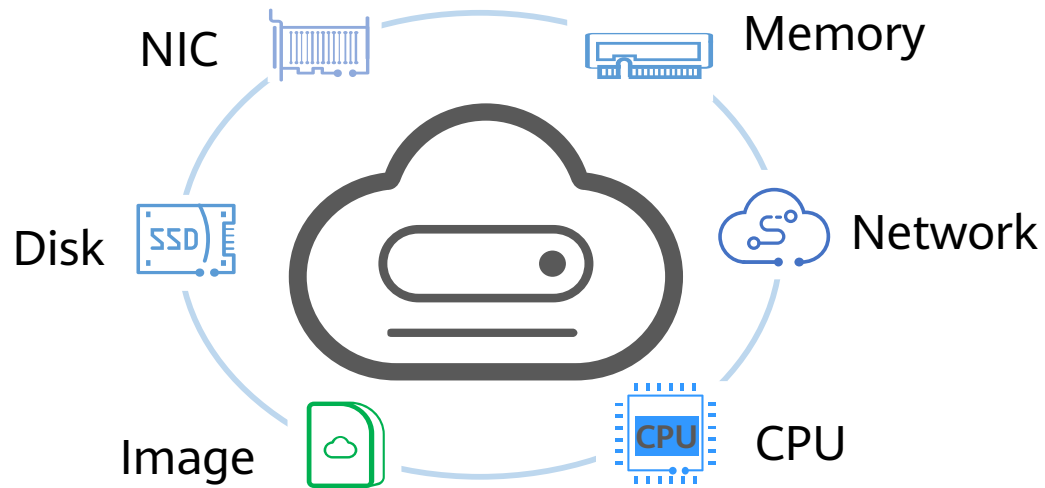
- 1. Typical Huawei Cloud Compute Services**
2. Five Principles of Compute Solution Design

Overview of Huawei Cloud Compute Services



ECS Positioning

Elastic Cloud Server (ECS) is a scalable, on-demand computing cloud server for secure, flexible, and efficient applications. It ensures the stable and uninterrupted operations of services.



- Elastic: Specifications are diverse and flexible.
- Computing-focused: Servers are treated as compute resources.

Instance Flavor

C6.8xLarge.4

1 2 3 4

- 1 Instance type
For example, s indicates general-purpose, c indicates computing, and m indicates memory-optimized.
- 2 Instance generation
For example, the 1 in s1 means the instance is a first-generation of instance, or the 6 in c6 means sixth-generation.
Generally, a larger number indicates a newer generation, which is more cost-effective.
- 3 Instance specifications
The number of vCPUs, for example, small, medium, large, xlarge, 2xlarge, 4xlarge, 8xlarge, and more.
- 4 Memory/vCPU ratio
The ratio of memory to vCPUs is expressed in a digit. For example, 4 indicates that there is 4 GiB of memory for each vCPU.

Instance Type

- General computing
- General computing-basic
- Large-memory
- Disk-intensive
- GPU-accelerated
- FPGA-accelerated
- Kunpeng memory-optimized
- Kunpeng ultra-high I/O
- ...



Tips for Selecting an Instance Type

Select an instance type based on the module features.

- Evaluate resource usage and select proper specifications.

1

2

Do not worry if you select an inappropriate instance type. You can:

- Change the instance type easily.
- Purchase a new ECS.

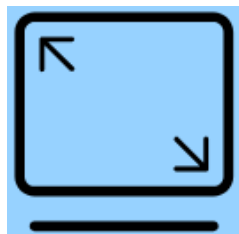
3

Optimizing instance types is not a one-time job. You can:

- Optimize instance types based on monitoring metrics.
- Check whether all key resources are being fully utilized.
- Change the instance type if resources are being wasted.
- Keep monitoring and find more opportunities for optimization.

Discussion: Instance Selection for Gaming Applications

Instances are not selected based on a specific industry or application. The selection is based on the components running on the instance.



Front-end service



Video game streaming service



Database service



AI-assisted module to support a personalized gaming experience

C7

General computing-plus

2-128 vCPUs
Maximum PPS: 12 million
Intel Xeon SkyLake

- ✓ High-performance websites
- ✓ Heavy- and medium-load enterprise applications

G6v/P2vs

GPU-accelerated

NVIDIA T4
8.1 TFLOPS single-precision floating-point computing

- ✓ G series: 3D animation rendering
- ✓ P series: deep learning and scientific computing

I7

Ultra-high I/O

Maximum PPS: 8 million
40 Gbit/s intranet bandwidth

- ✓ High-performance relational databases
- ✓ NoSQL databases
- ✓ Elasticsearch

Ai1

AI-accelerated

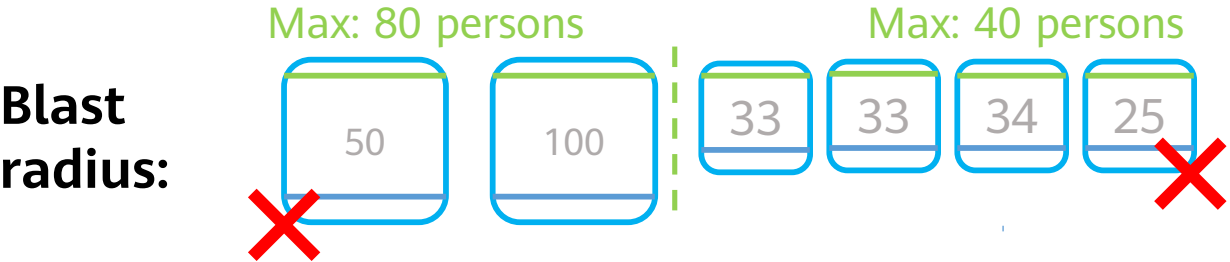
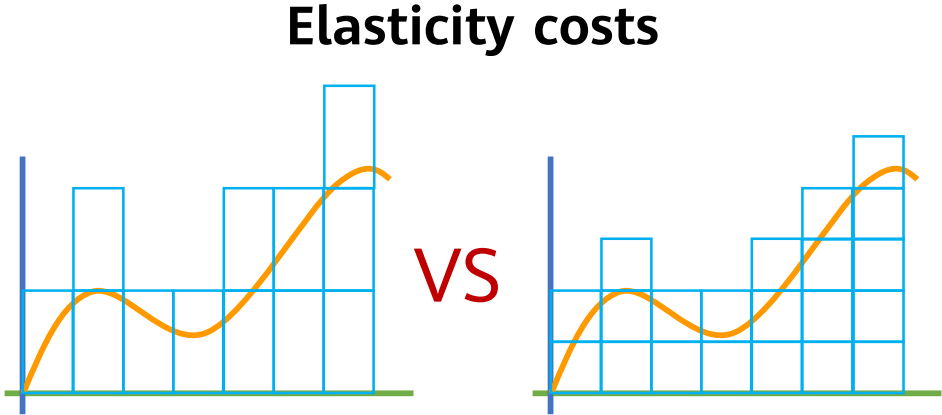
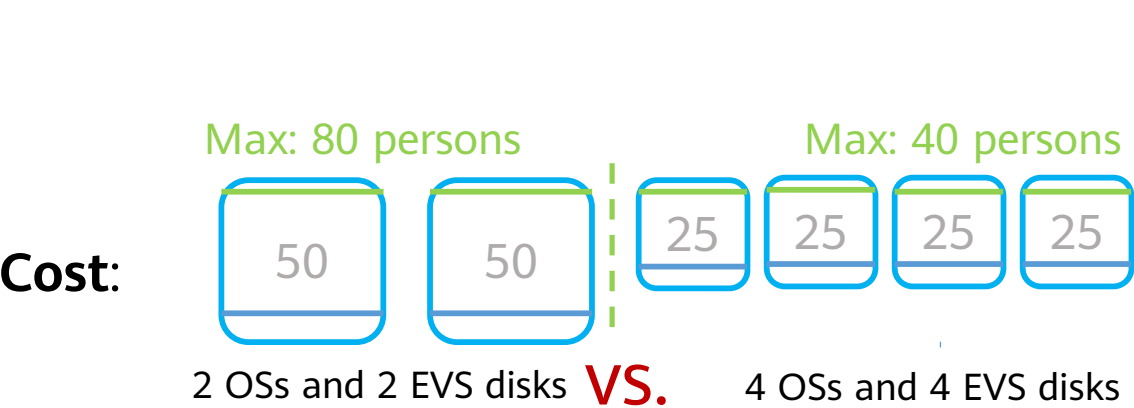
1-16 Ascend 310
Designed based on the Atlas 300I accelerator card

- ✓ Machine vision
- ✓ Speech recognition
- ✓ Natural language processing

Which Specifications Should I Select?



Comparison of ECS Specifications



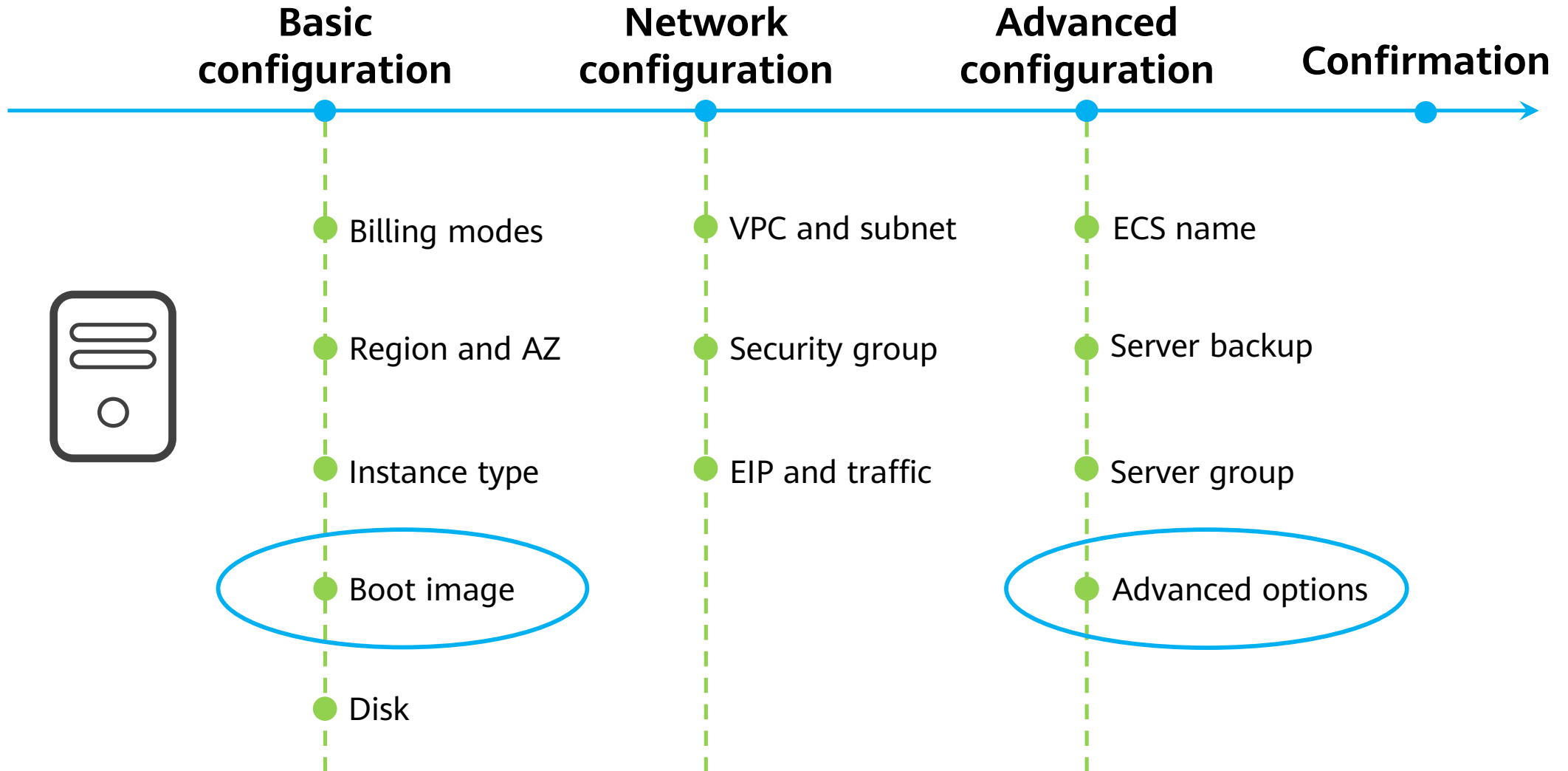
- The blast radius of an incident with multiple low-specification servers is smaller.
- It costs less to add multiple low-specification servers.

Select specifications based on your service scale.

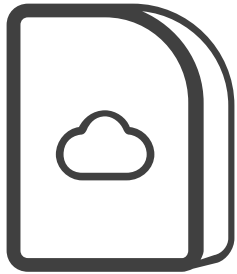
How Do I Initialize an ECS?



Initialize an ECS



Initialize an ECS Using an Image



IMS

- An image is a template of an ECS environment.
- An image contains VM resource type restrictions.
- An image contains an OS.
- An image can be used to create identical ECSs.
- A variety of image types are provided by Huawei Cloud.

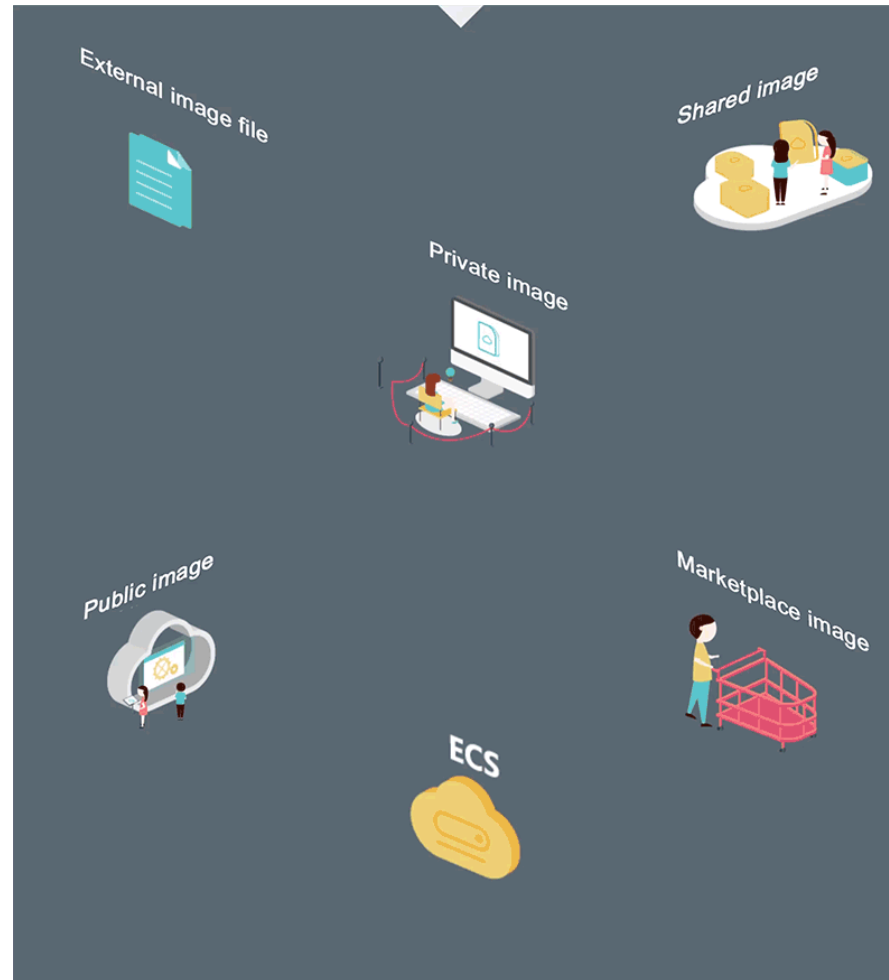
Image Types and Features

Public images

- All public images are thoroughly tested before being published.
- Operating systems are pre-installed in public images.
- Huawei Cloud periodically updates and maintains public images.
- A public image is a stable environment for customization.

Private images

- You can create a private image from your ECS.
- You are responsible for the image security.
- When you create an image, your ECS should be stopped.
- You can install custom software as needed.



Marketplace images

- Marketplace images are third-party images available in KooGallery.
- All marketplace images are thoroughly tested before being published.
- The billing is determined by image providers.
- Specific applications are pre-installed in marketplace images.

Shared images

- You can share private images with other accounts.
- You need to configure permissions before sharing.
- Image providers are responsible for the security of shared images.
- Ensure that a shared image is from a trusted sharer.

Initialize an ECS Using Scripts



Leverage user data



- User data is the data interaction between users and servers.
- User data in a specific format can be regarded as a script.
- Supported formats include Bash, PowerShell, Batch, and more.

Size



- Maximum size: 32 KB
- You are advised to keep larger scripts in OBS and then download and run them from there.

Execute once



- The execute once option only occurs when the server is started for the first time after being purchased.
- It does not take effect to modify the user data of a started server.

Required execution time



- Before the execution is complete, the server is started.
- You can log in to the server before script execution is complete.

Initialize an ECS Using Scripts

Advanced Options ☒ Configure now

User Data **As text** As file [Learn how to inject user data.](#)

```
#!/bin/bash
cd /root
yum install -y httpd php php-fpm php-server php-mysql
```

416/32,768

Tag It is recommended that you use TMS's predefined tag function to add the same tag to different cloud resources. [View Predefined Tags](#)

Tag key Tag value

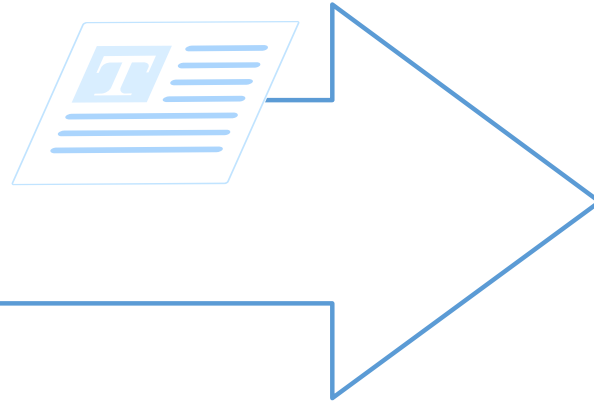
You can add 10 more tags.

Agency --Select-- [Create Agency](#) ?

When purchasing an ECS, if **Login Mode** is set to **Key pair** or **Set password later**, you can inject user-defined data in text or file format in **Advanced Options**. You can use scripts to initialize the server.

Initialize an ECS Using an Image or Scripts

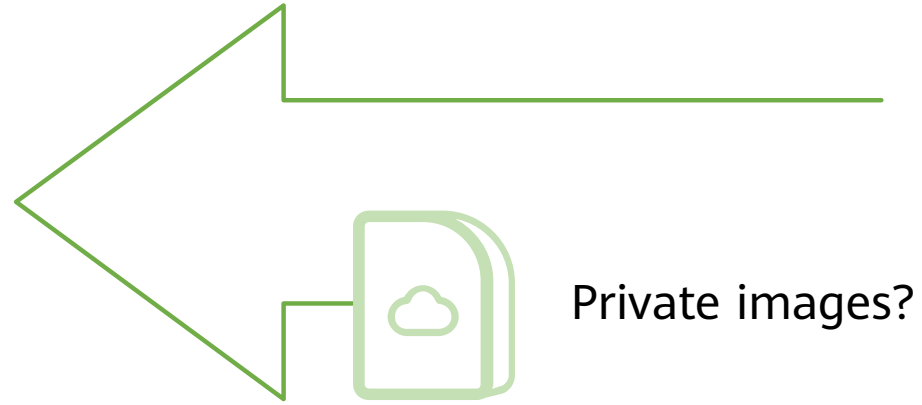
Initialization scripts?



A server must:

work upon startup

VS



Private images?

Advantages and Disadvantages of Private Images and Initialization Scripts



Advantages:

- Fast startup, only a private image is required.
- Easy to use



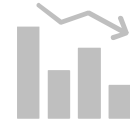
Disadvantages

- Slow image update
- High cost. Private images storage is charged.
- Creating an image is a demanding job.
- Historical images need to be managed.



Advantages:

- Flexible, almost real-time updates
 - Integrated with OBS, effective for new servers
- Stable, controllable, and changes traceable

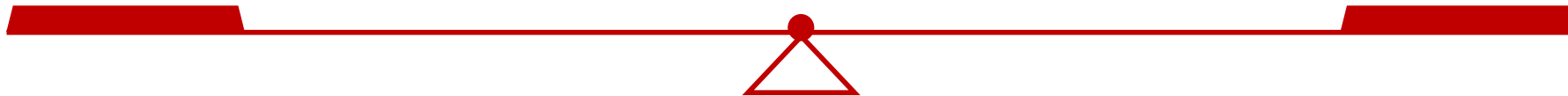


Disadvantages

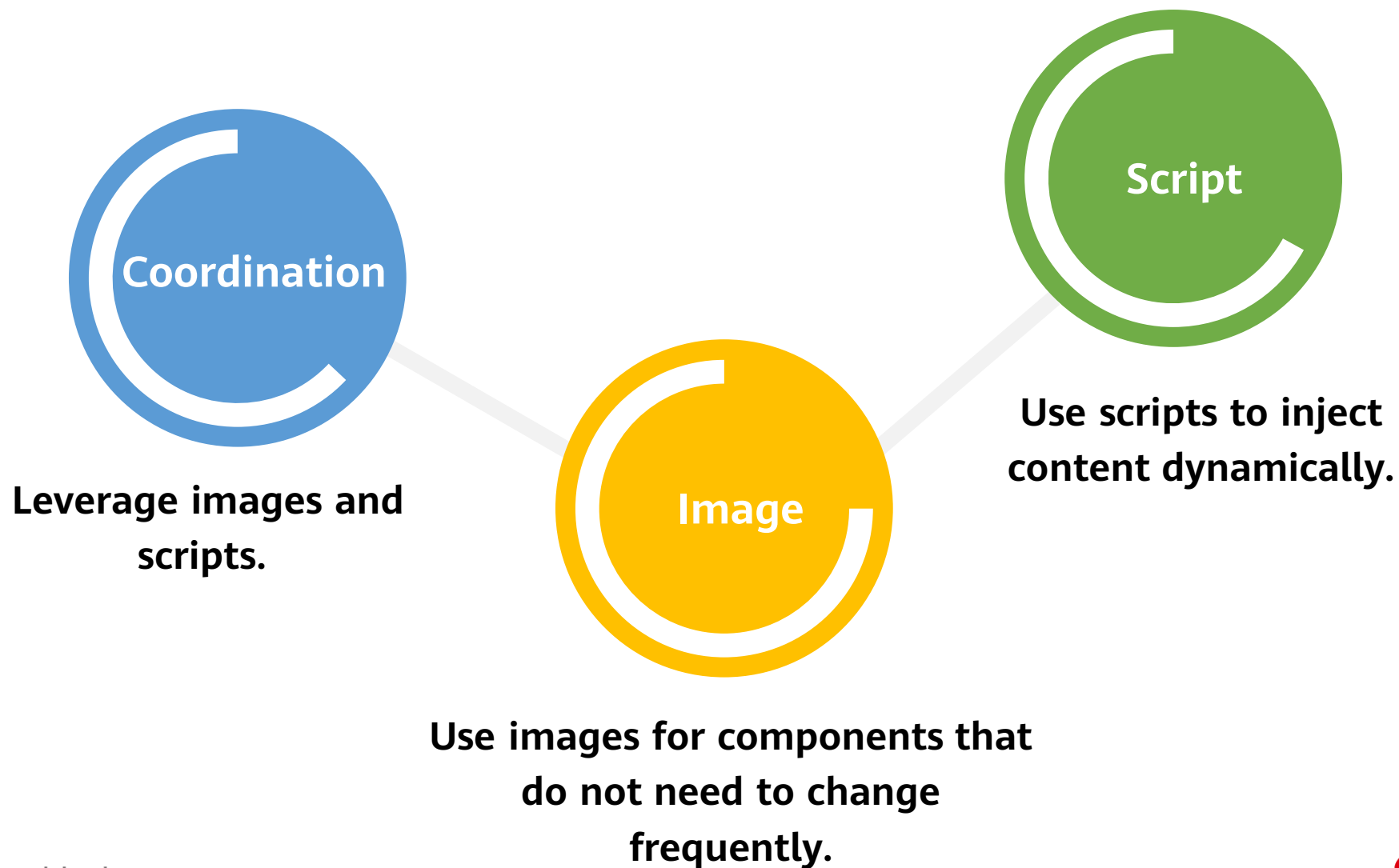
- Slow execution of complex scripts
- High technical requirements for writing scripts

Private images

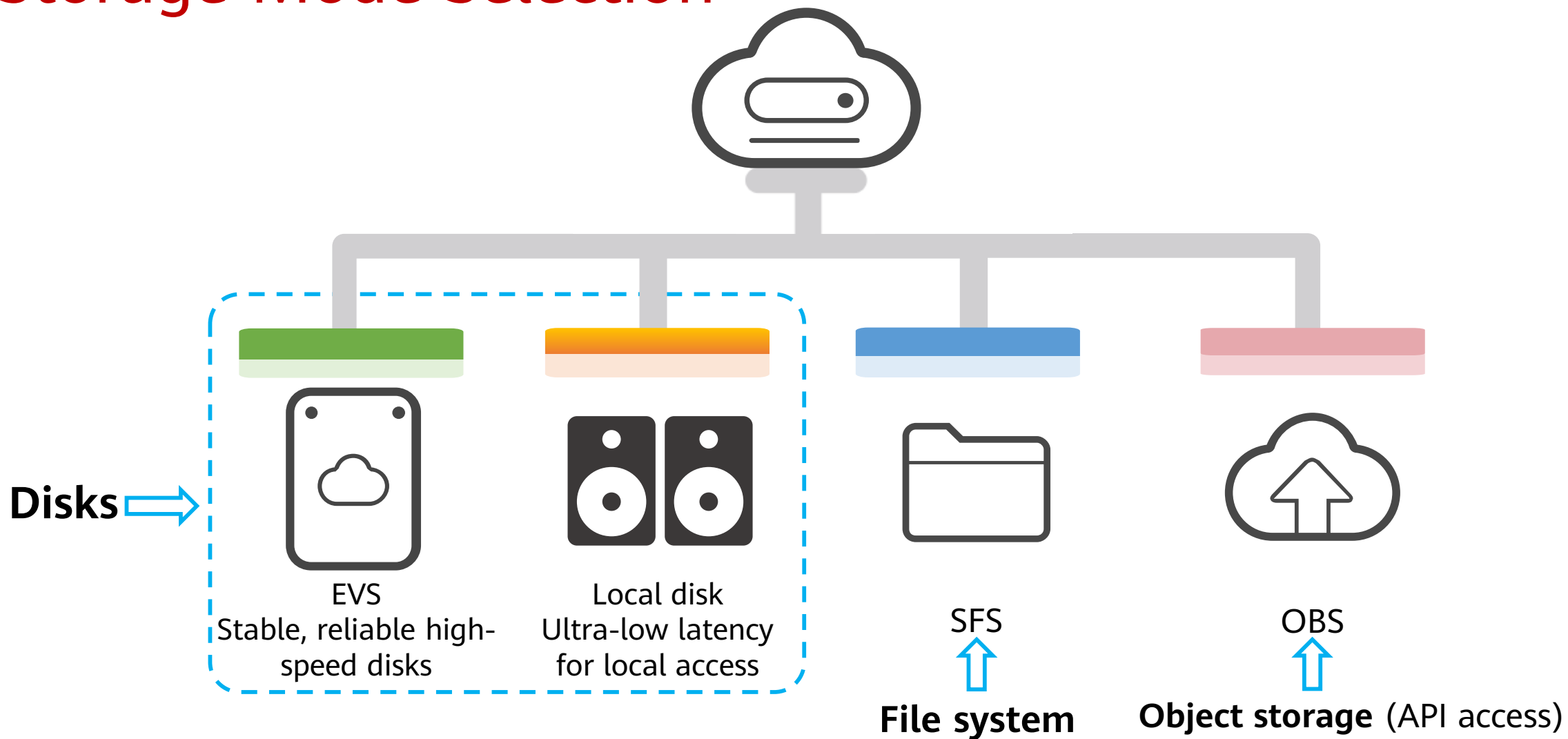
Initialization scripts



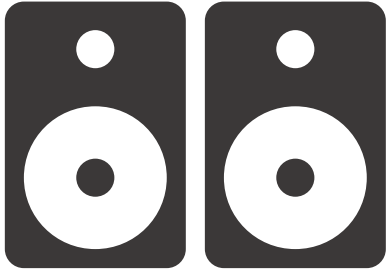
Selection Principles



Storage Mode Selection



Features of ECS Local Disks



ECS local disks

- Ultra-low latency
- Super high IOPS
- Available only for some types of instances, such as i3 and d6
- Only instances are billed. All is included in the instance expenditure.
- Less data reliability

Application scenario: Store data that is not that important or that can be restored.

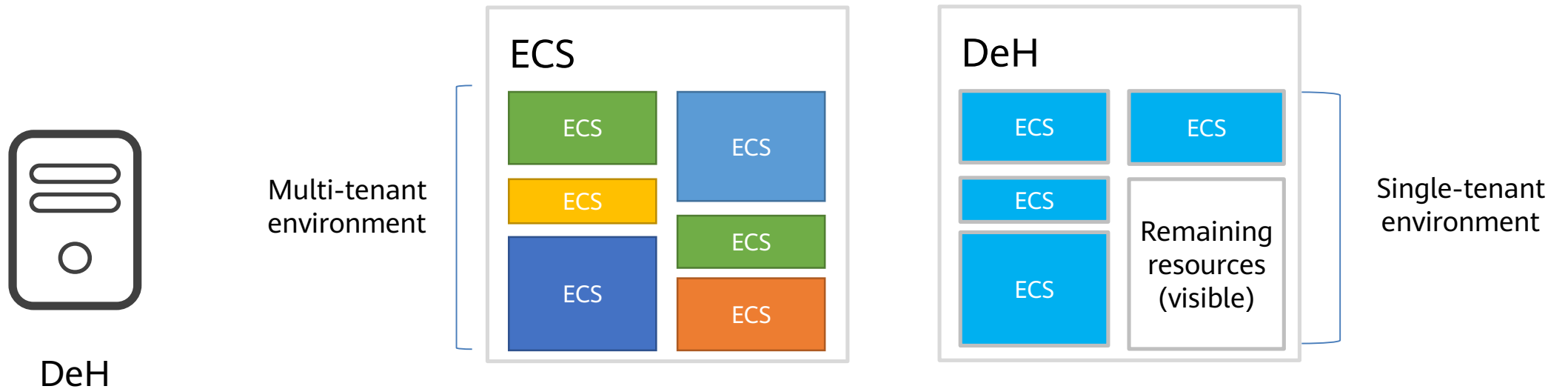
For example, ECS local disks can store swap files or be used as cache.

Comparison Between Storage Solutions

Item	EVS	Local Disk	SFS	OBS
Working model	Block storage, which is used as disks		File storage, which is accessed through the network file system	Object storage, which is accessed via APIs
Reliability	Multiple copies in a single AZ Reliability: 9 nines reliability (99.9999999%)	No high availability design and data is lost if server is stopped	Reliability: 10 nines (99.99999999%)	Multiple copies in multiple AZs Reliability: 12 nines (99.999999999%)
Security control	EVS encryption	OS encryption	SFS encryption	OBS encryption
Performance	Millisecond latency and high bandwidth	Ultra-low latency for local access	3–20 ms latency	10 ms latency, high concurrency, and high throughput
Storage space limit	Up to 32 TB for a single disk, pre-purchase	Depending on the instance type	No limit (depending on the type)	No limit on quantity and space, and up to 48 TB for a single object
Cost	High. EVS disks must be used with servers.	Free. The cost is included in the instance expenditure.	Different types of file systems are provided.	Low. Different storage classes are available.
Access from Internet	No. EVS disks must be attached to servers for use.		No. An SFS file system must be mounted to servers and given the specific network address for access.	Yes. Public access requires corresponding permissions.
Application scenario	Workloads running on traditional disks		File sharing, media processing, and content management	Large-scale data storage, important data sources for big data, archive data storage, and data distribution on the Internet

DeH

A Dedicated Host (DeH) is a physical server fully dedicated for your own services. DeH allows you to ensure performance by keeping compute resources isolated.



Note: ECSs in different colors belong to different tenants.

Users can view the resource usage on a DeH and dynamically adjust the distribution of VMs on the DeH.

DeH Features



Exclusive

- Fully dedicated, DeH hosts only your ECSs, ensuring high computing performance and high stability.



Secure

- Keep your services secure with host-level compute isolation, making it easier to meet compliance requirements.



Flexible

- Scale up ECSs deployed on a DeH based on your service requirements to improve resource utilization.

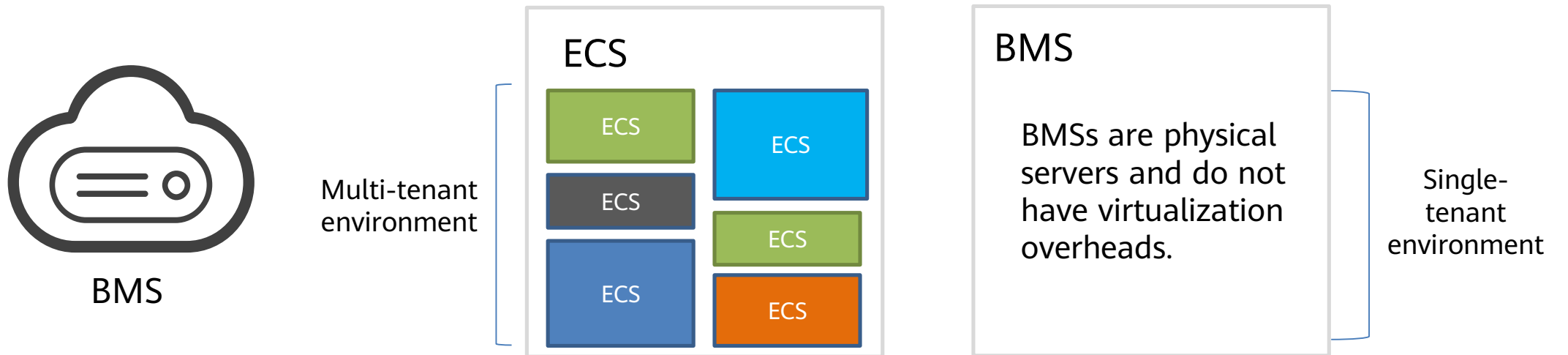


Cost-effective

- Use existing server-bound software licenses. You can take advantage of these existing investments to keep costs down.

BMS

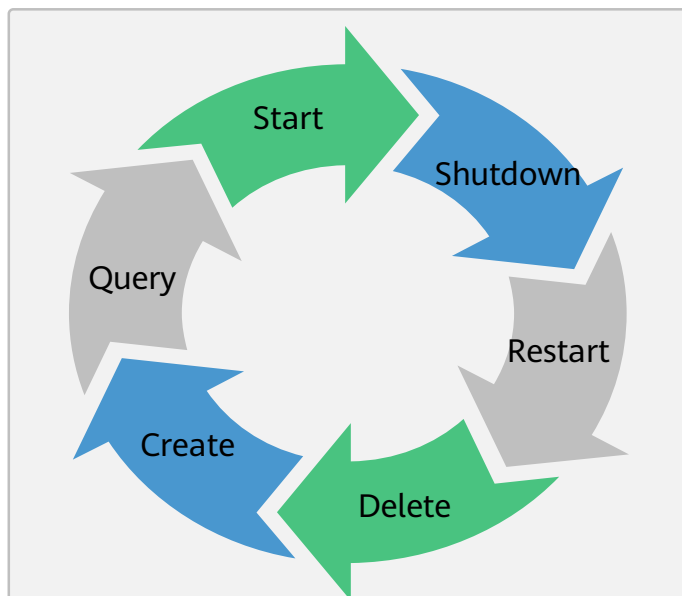
Bare Metal Server (BMS) provides dedicated physical servers in single-tenant environments. It features both the scalability of Elastic Cloud Servers (ECSs) and stability of physical servers, delivering the performance and security required by core applications.



BMS Functions

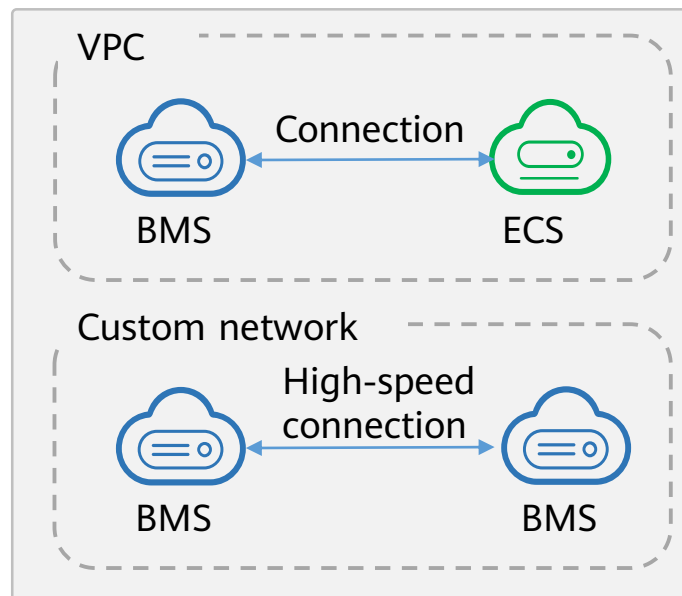
Provisioning within 5 minutes

Self-service purchase through just a few clicks, BMS lifecycle management



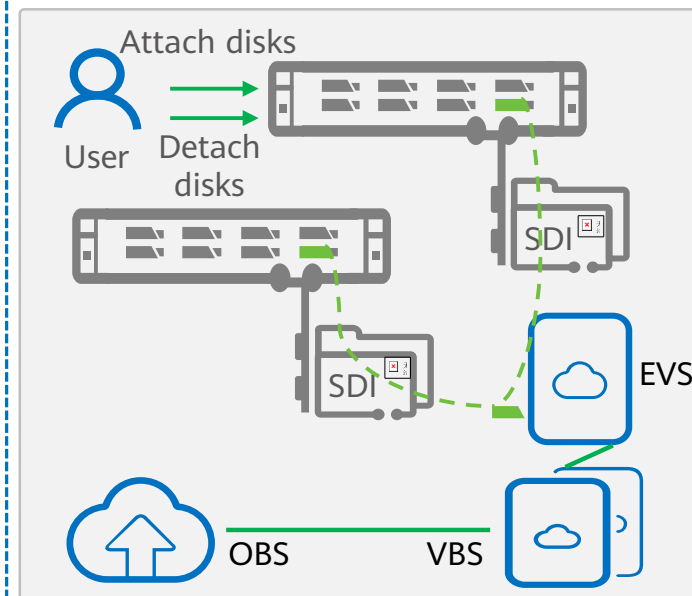
Communication with ECSs

BMSs can communicate with ECSs in VPCs. Custom networks enable high-speed communications between BMSs.



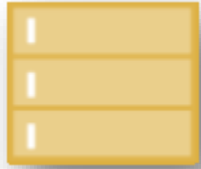
Shared EVS disks

With the exclusive SDI technology, shared EVS disks can be attached to BMSs for the cloud deployment of applications that require shared storage, such as Oracle RAC.



Application Scenarios of BMS

Scenarios



Core databases



High-performance computing



High security



Big data

Examples

- | | | | |
|---|---|--|--|
| <ul style="list-style-type: none">• Enterprise development and testing environments• Databases• Typical Oracle database | <ul style="list-style-type: none">• Supercomputing center• Biopharma• Genetic engineering | <ul style="list-style-type: none">• Securities• Government and enterprise finance• Insurance | <ul style="list-style-type: none">• Mass storage• Massive data analysis |
|---|---|--|--|

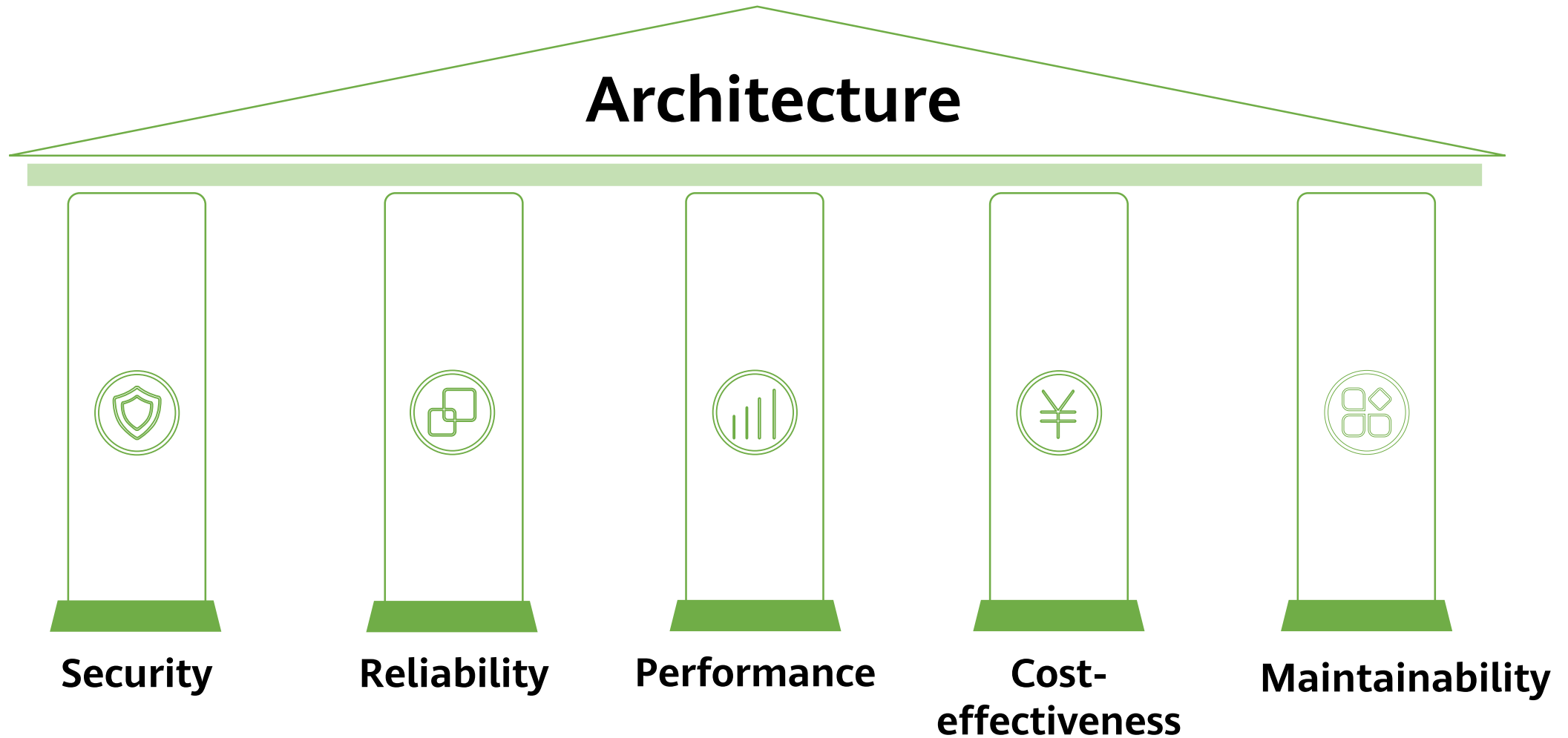


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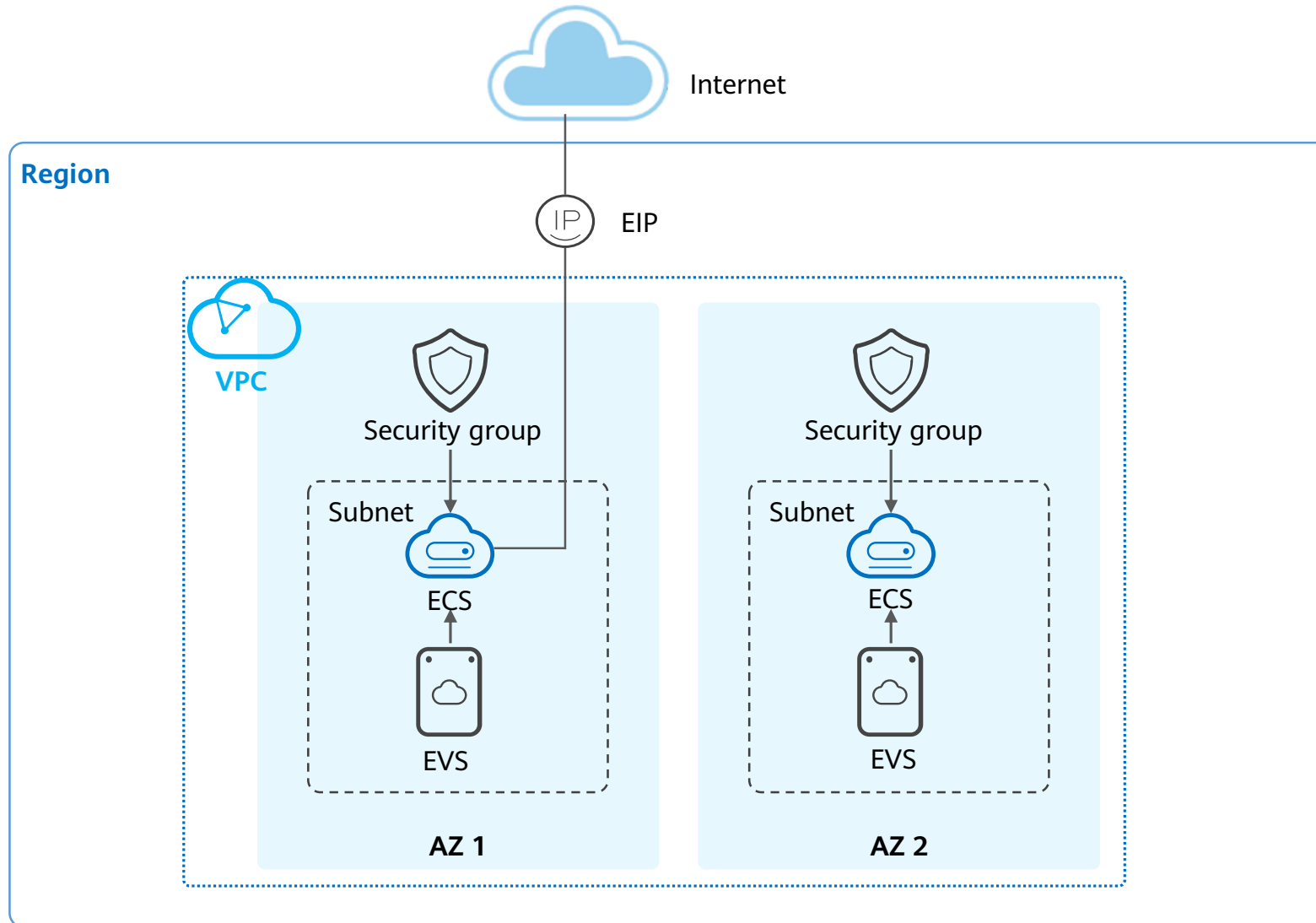
1. Typical Huawei Cloud Compute Services

2. Five Principles of Compute Solution Design

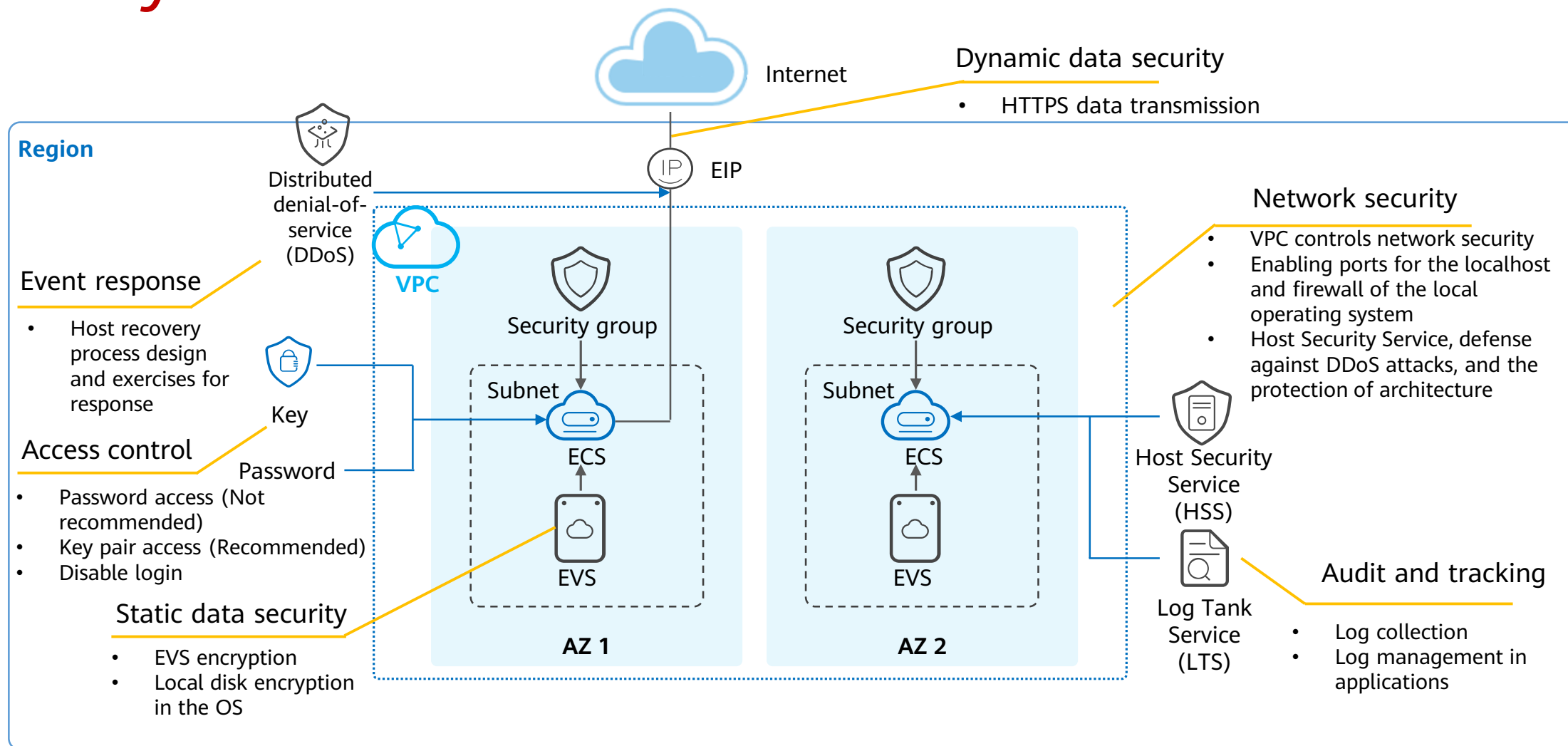
Five Principles of Computing Architecture Design



Example Architecture

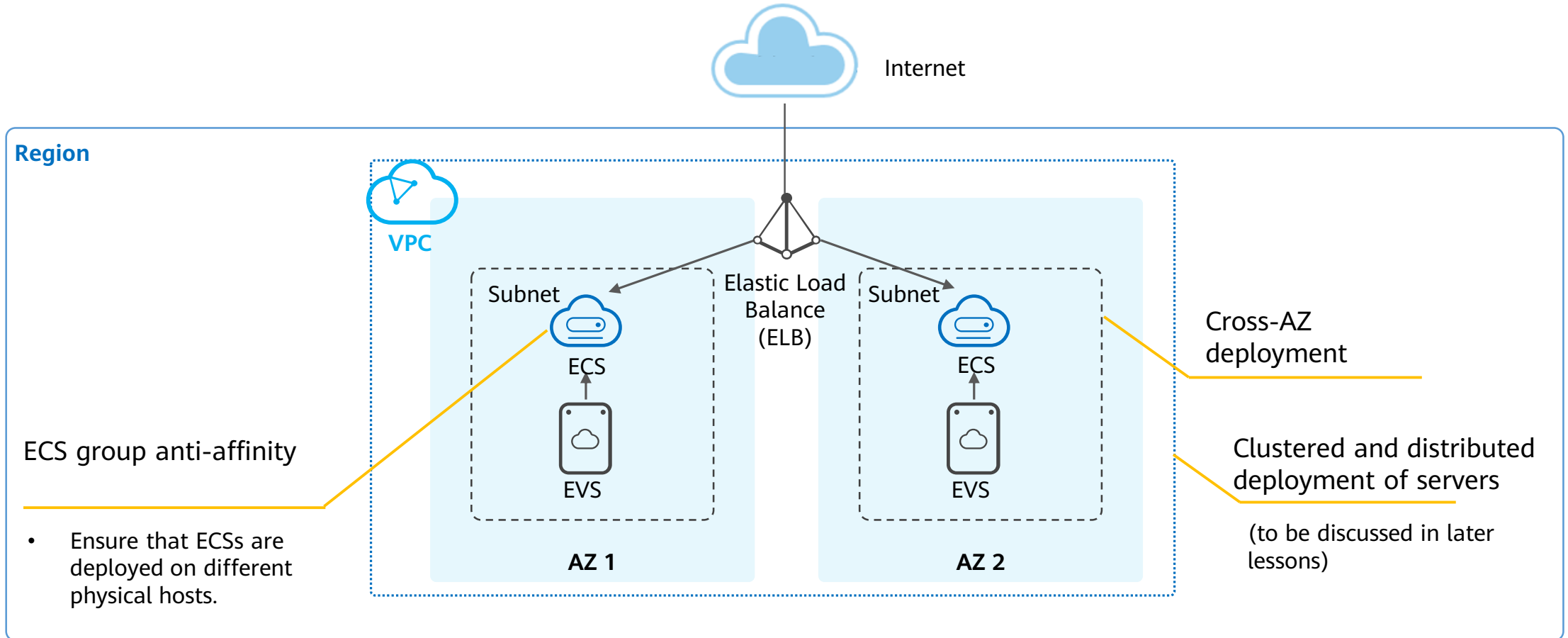


Security

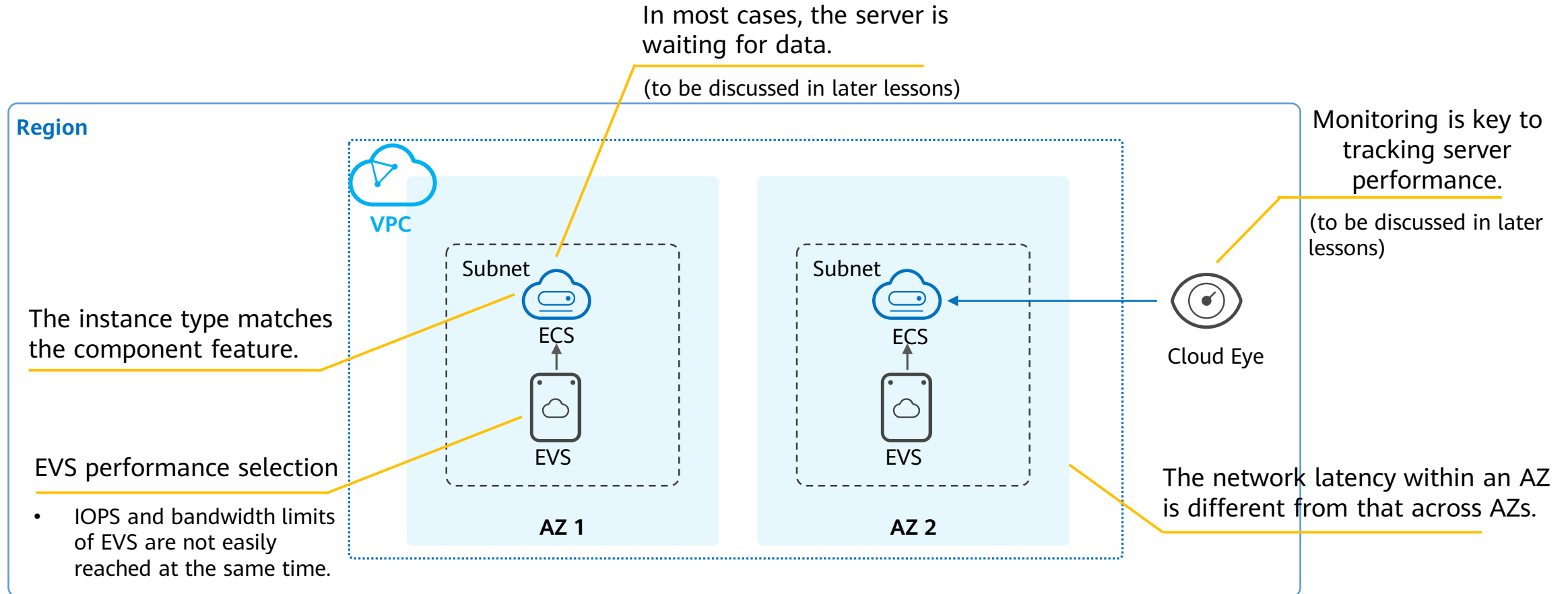


Reliability

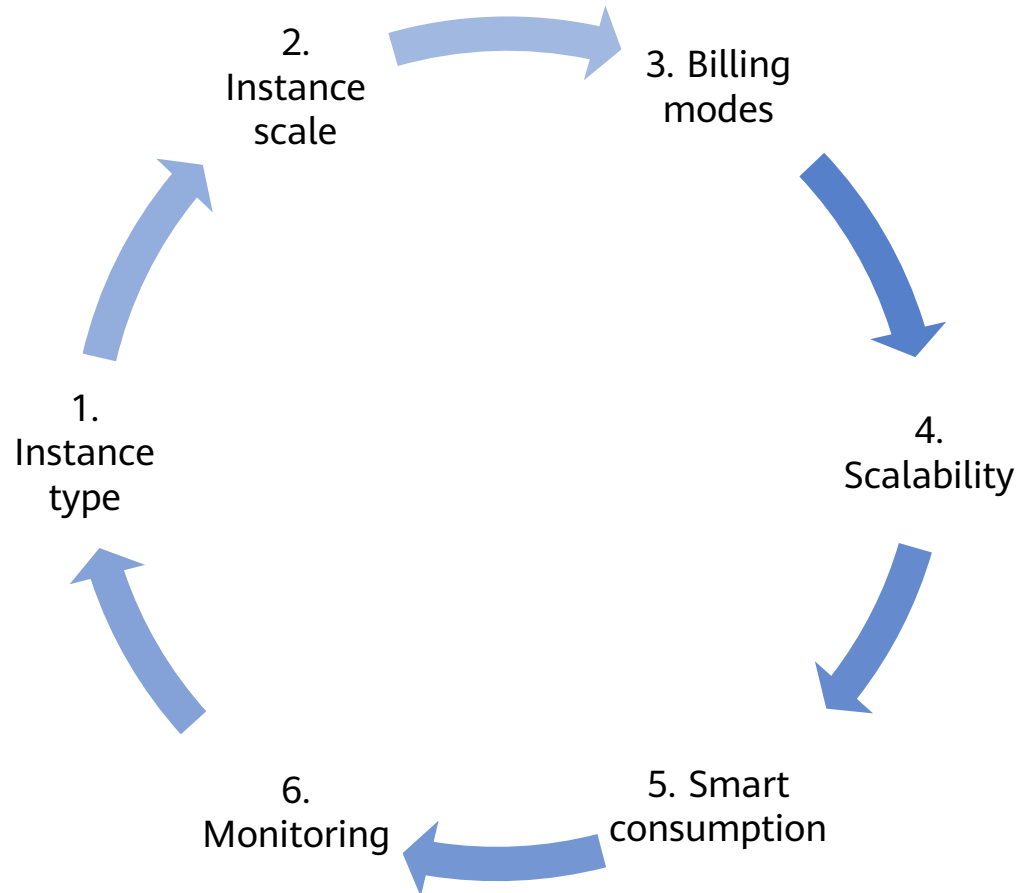
- An ECS is deployed on a single node. You need to consider solution reliability.



Performance



Costs



Pricing details: <https://www.huaweicloud.com/intl/en-us/pricing/calculator.html#/ecs>

1. Instance type

- Select instance types based on the features of components running on the server.

2. Instance scale

- Select an instance that can make the most of host resources.

3. Billing modes

- Pay-per-use, yearly/monthly, and spot pricing
- Billed by traffic, inbound traffic is free, and outbound traffic from the private network to the same region is free.
- Balance the traffic and compute.

4. Scalability

- Estimate traffic changes and set scaling policies.

5. Smart consumption

- Stop the ECS when it is not in use. (If EVS disks are not detached and the EIP is not released, they will continue to be billed. Billing only stops for the ECS.)

6. Monitoring

- Keep monitoring resource usage and continuously optimize the settings accordingly.

O&M

1. ECS O&M
2. Operating system patches
3. Application software upgrades
4. Self-built applications upgrades
5. Data backup and host backups
6. Other security-related maintenance described previously

Quiz

(Multiple-choice question) Which of the following statements about using scripts to initialize a server are true?

- A. Various types of scripts can be used to initialize a server, such as Bash and PowerShell.
- B. If the scripts are too big, you can store them in an OBS bucket and inject them as files that can be download and run after a server is created.
- C. To update the scripts, you can modify the scripts on the started server.
- D. Before the execution is complete, you can log in to the server.



Summary

- This lesson described Elastic Cloud Server (ECS) features, instance selection, initialization mode selection, and storage solution selection, as well as the definitions, features, and application scenarios of Bare Metal Server (BMS) and Dedicated Host (DeH). This lesson also discussed how to provide better compute services from the perspective of the five principles.



Acronyms and Abbreviations

- AZ: Availability Zone
- ECS: Elastic Cloud Server
- BMS: Bare Metal Server
- BYOL: Bring Your Own License
- DeH: Dedicated Host
- EVS: Elastic Volume Service
- OBS: Object Storage Service
- SFS: Scalable File Service
- EIP: Elastic IP
- DDoS: Distributed Denial-of-Service
- HSS: Host Security Service
- LTS: Log Tank Service
- VPC: Virtual Private Cloud

Thank You.

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